Passive Infrared (PIR) Sensor



What is it?

- PIR sensors detect infrared radiation emitted by human bodies.
- They are often used in motion detection systems, such as security alarms and automatic lights.
- PIR sensors have a wide detection angle and can detect motion from a distance.
- They are relatively inexpensive and easy to install.
- PIR sensors can be affected by changes in temperature or other environmental factors.

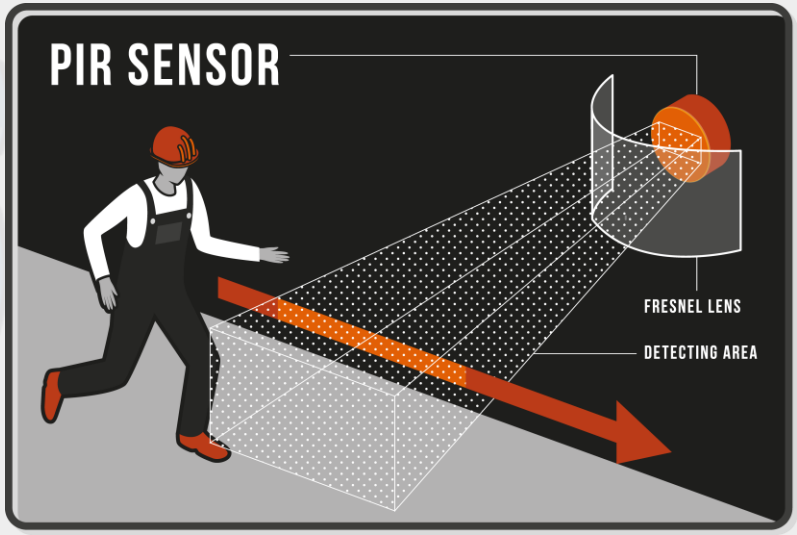
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* Image courtesy: Vector Stock

Working Principle

How it works?

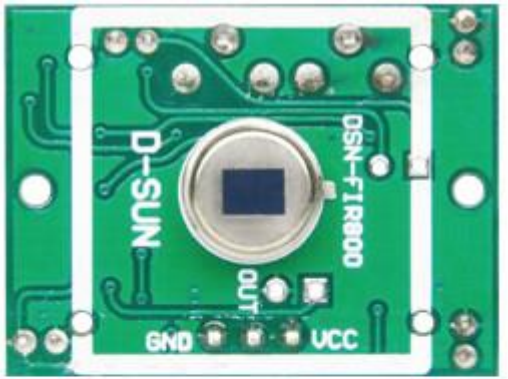
- All objects, including the human body, at temperatures above absolute zero (0 Kelvin, -273.15 °C) emit heat energy in the form of infrared radiation.
- The hotter an object is, the more radiation it emits.
- This radiation is not visible to the human eye because it is emitted at infrared wavelengths.
- The PIR sensor is specifically designed to detect such levels of infrared radiation.



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* Image courtesy: https://www.contintechind.com/pir-sensor/

HC-SR501 PIR Sensor





Main Two Elements:

- A **pyroelectric sensor**, which you can see in the image below as a round metal with a rectangular crystal in the center.
- A special lens called a **fresnel lens** which Focuses the infrared signals on the pyroelectric sensor.

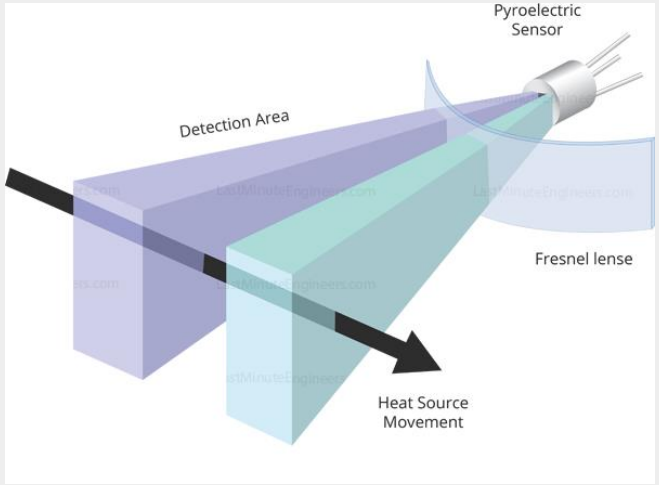
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* Image courtesy: lastminuteengineers.com

HC-SR501 PIR Sensor

But how?

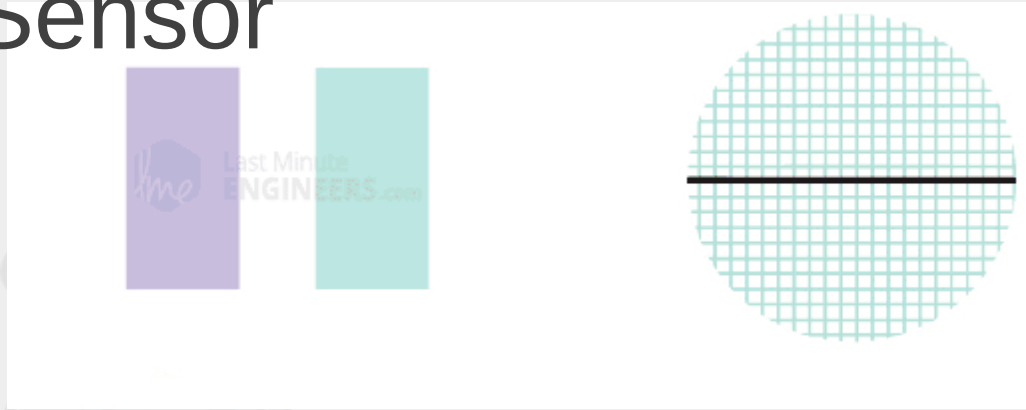
- **Pyroelectric sensor:** Two infrared sensor electrodes behind a window.
- **Window:** Allows infrared radiation to pass through.
- **Electrodes:** Detect changes in infrared levels.
- **Wiring:** Cancels out ambient infrared levels.
- **Output:** Generated when there's a difference in infrared levels between the electrodes.



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* Image courtesy: lastminuteengineers.com

HC-SR501 PIR Sensor



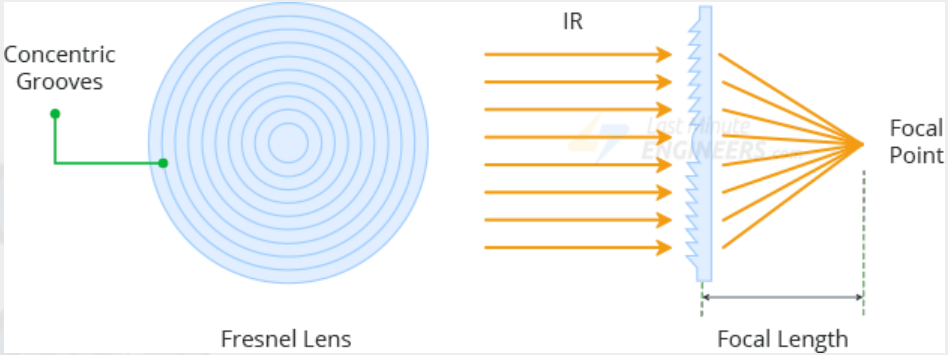
But how?

- When there is no movement around the sensor, both slots detect the same amount of infrared radiation, resulting in a zero output signal.
- But when a warm body passes by, it first intercepts half of the sensor. This causes a positive differential change between the two halves. When the warm body intercepts the other half of the sensor, the sensor produces a negative differential change.

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* Image courtesy: lastminuteengineers.com

HC-SR501 PIR Sensor



The Fresnel Lens:

- It increases the range and field of view of the PIR sensor. Its light gathering capability make it extremely useful for making PIRs small in size yet powerful.
- A Fresnel lens consists of a series of concentric grooves carved into the plastic. These contours act as individual refracting surfaces, gathering parallel light rays at a focal point. As a result a Fresnel lens, although smaller in size, is able to focus light similarly to a conventional optical lens.

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* Image courtesy: lastminuteengineers.com

HC-SR501 PIR Sensor

Specifications:

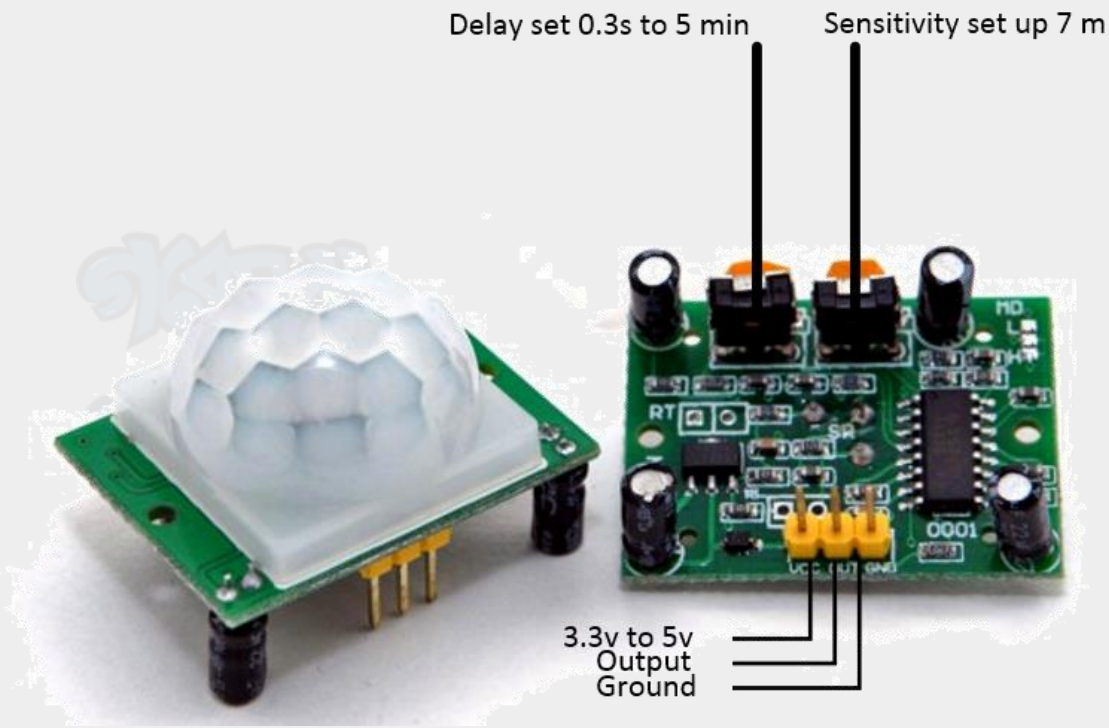
- Based on BISS0001 PIR Controller.
- 3.3v Operation. On board regulator inside, accepts 5-12v DC.
- Range: 3-7 Meter Max (can be adjusted).
- Direction angle: 120 degree.
- Pulse delay: 30ms- 5 minutes.
- Digital output.



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* Image courtesy: lastminuteengineers.com

Pinout



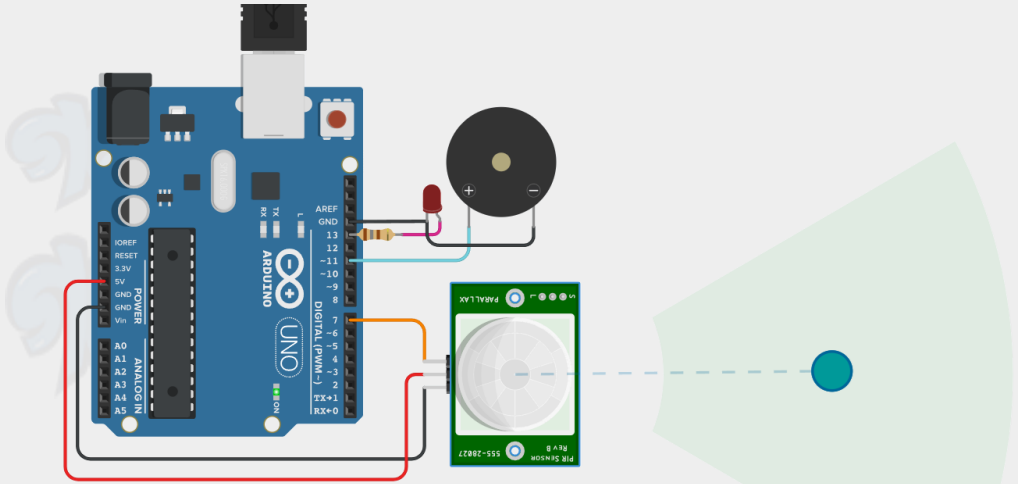
- VCC: 3.3/5V
- GND: Ground (0V)
- Out: Digital out (same voltage as VCC)

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* Image courtesy: https://shopee.ph

Sample Circuit

- 7: Input
- 11: Buzzer
- 13: LED
- GND: Ground (0V)
- VCC: 5V



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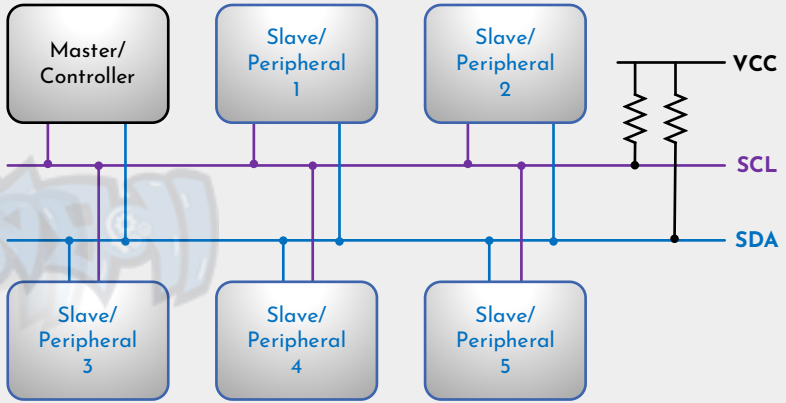
Sample Output



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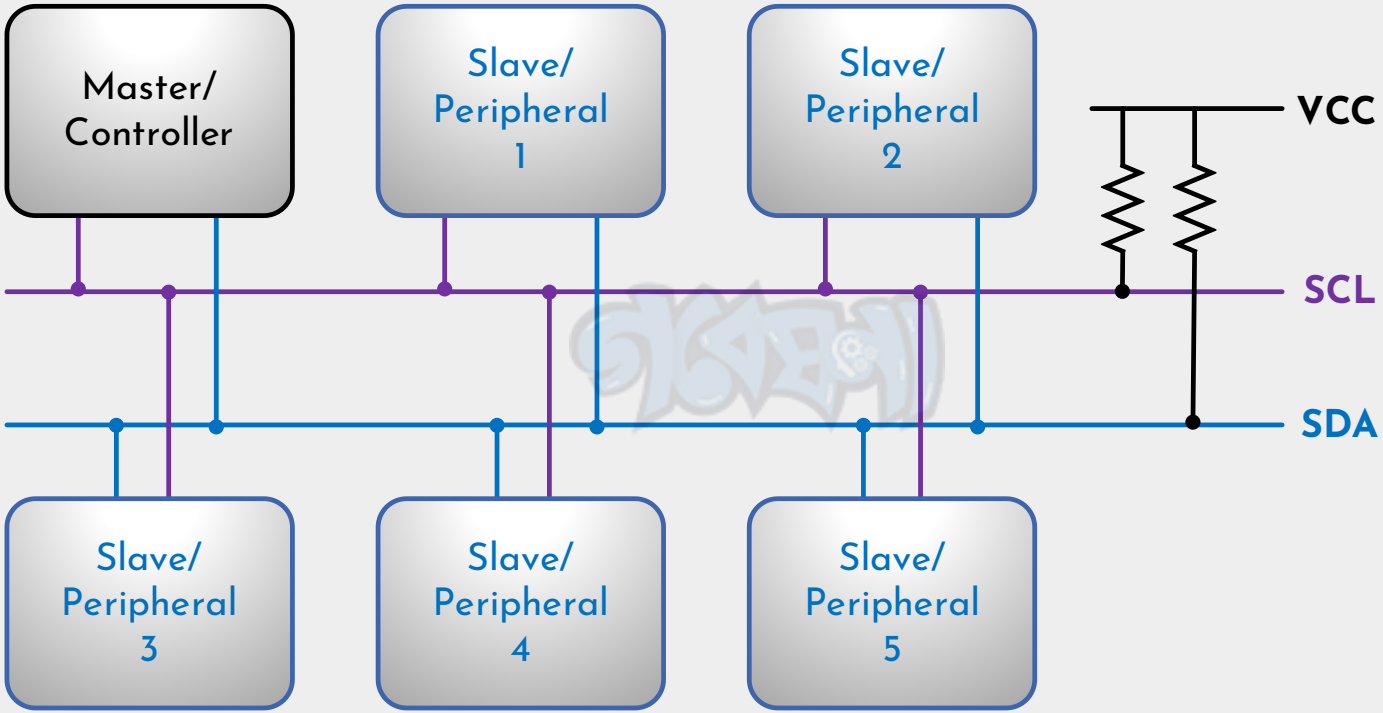
Inter Integrated Circuit Protocol

- Invented by Philips Semiconductor.
- Serial Communication Protocol
- 3 wires needed, (SCL, SDA, GND)
- 7 bit device address
- Synchronous protocol
- Master-Slave method
- Transfer rate: 100kbps-5Mbps



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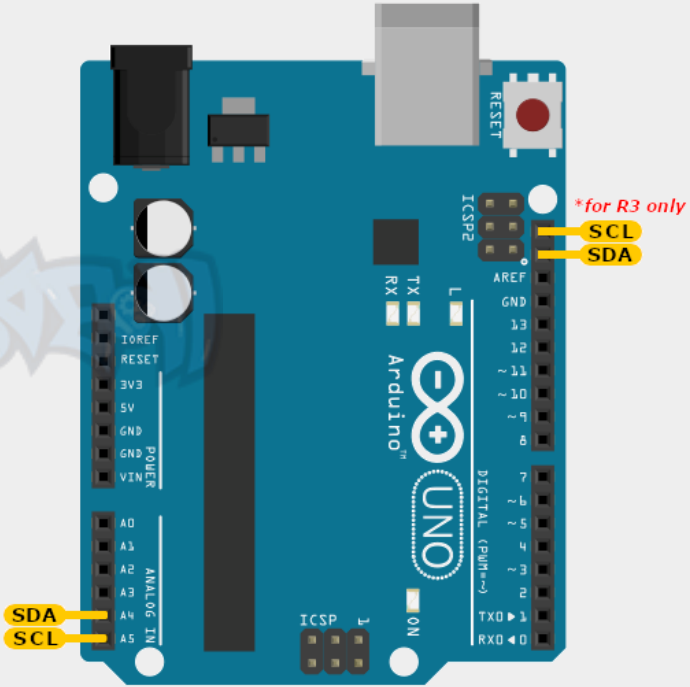
I²C Protocol



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Pinout for Arduino Uno R3

- SDA: Analog 4 (A4)
- SCL Analog 5 (A5)
- Voltage level: 5V



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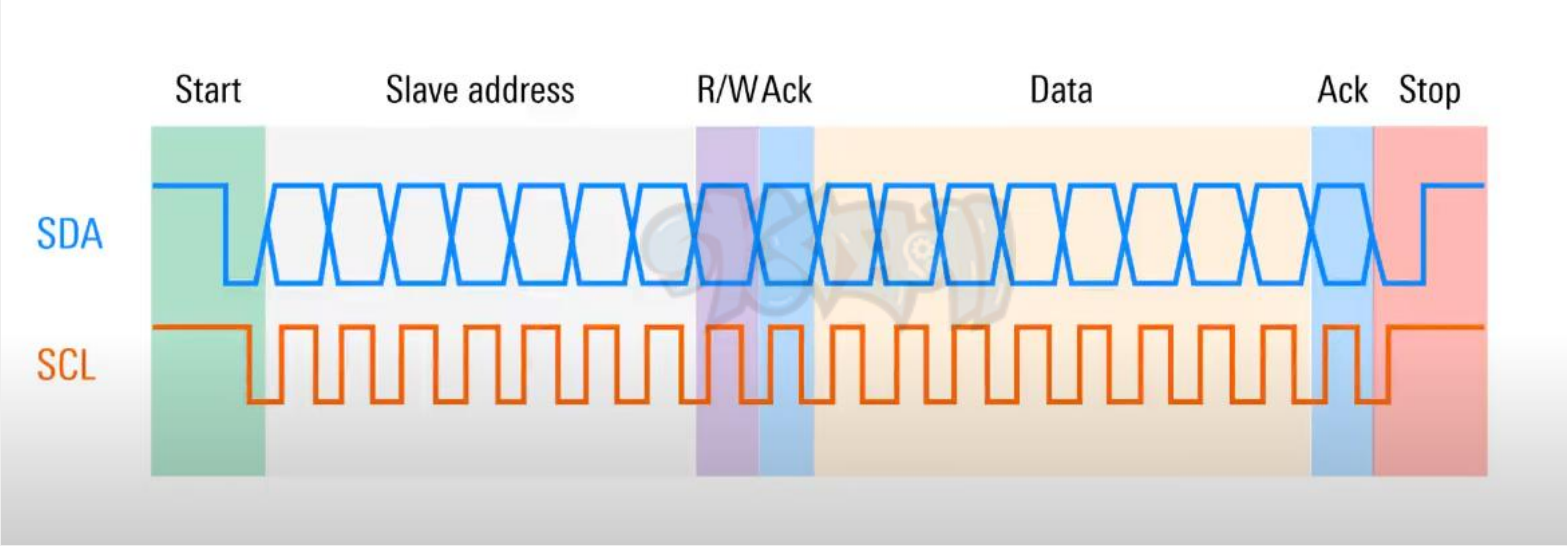
I²C Protocol



- Start: SDA high to low, SCL high to low
- Address: 7 Bit address with 1 bit R/W bit, total 8 bit
- Acknowledgement: Comes from the receiver. 0 means ACK, 1 means NAK
- Data: 1 byte (8 bit) per frame. MSB first.
- Stop: SCL low to high, SDA low to high.
- R/W: 1 means read, 0 means write

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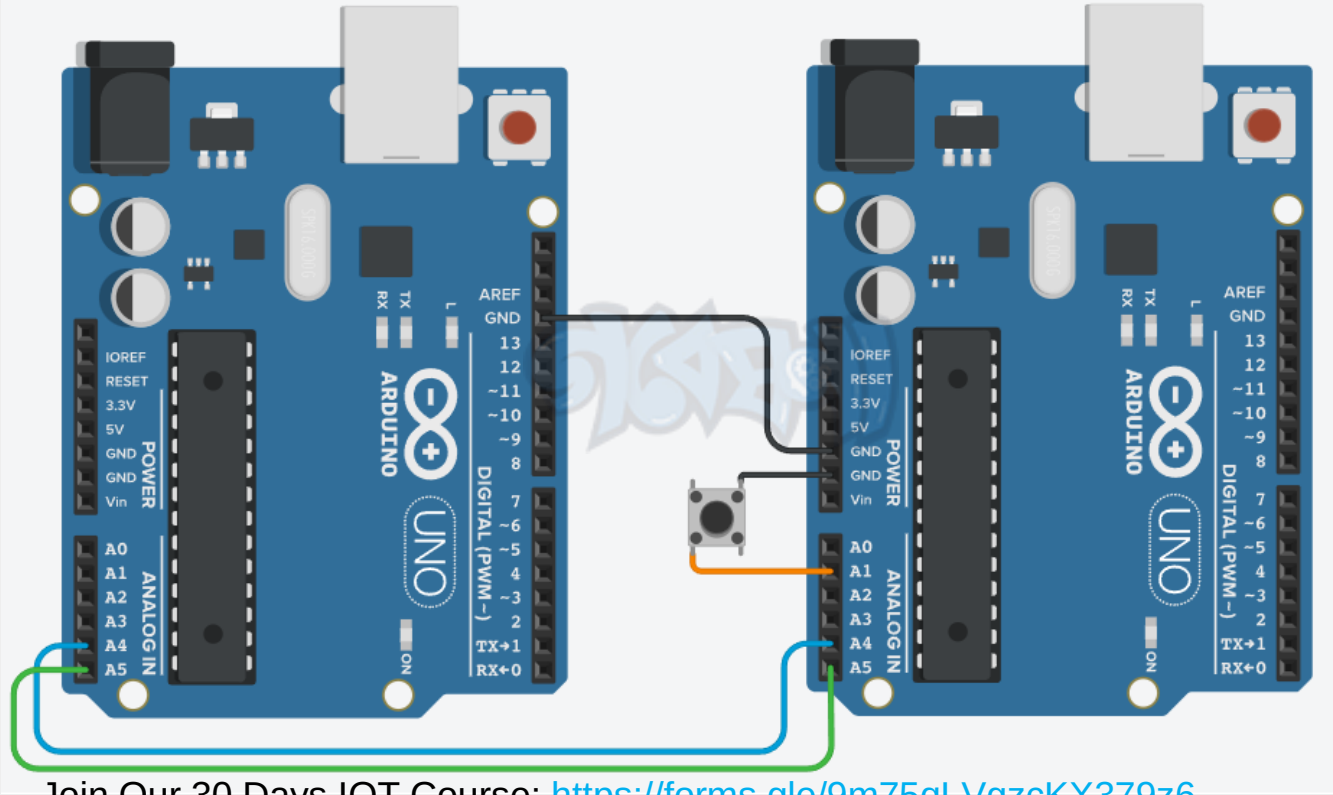
Timing Diagram



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Sample Circuit

Slave



Master

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Assignment

Write a code for Arduino as a slave to turn on/off led 13 with i2c protocol.

Use buttons and leds to simulate Master.

Slave address: 10 (0000 1010)

Turn on LED : 1 (0000 0001)

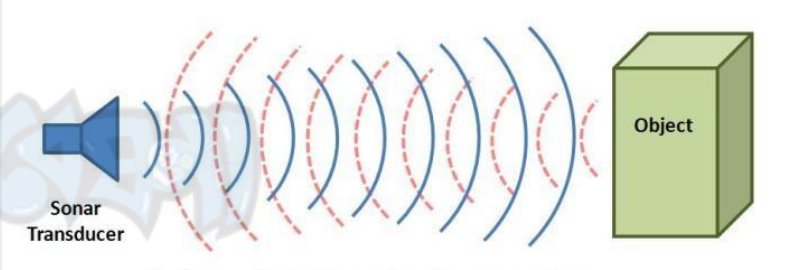
Turn off LED : 0 (0000 0000)

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Sonar Sensor

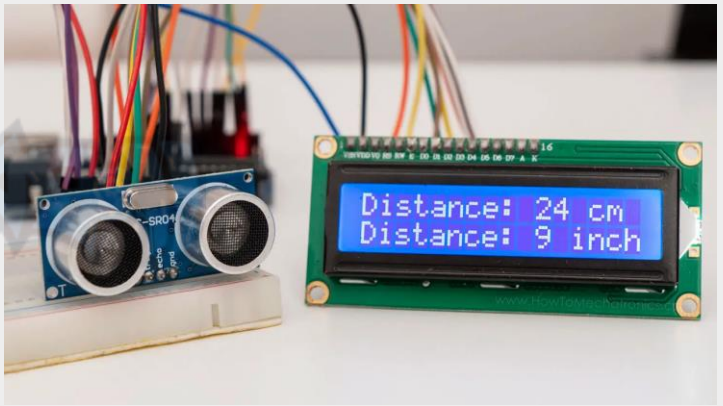
- Usage: Distance measurement
- Operated with ultrasound
- Frequency: 40,000+ Hz



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Sonar Sensor

- Usage: Distance measurement
- Operated in 5v
- Range: 2cm - 400cm
- 3/4 wires needed, (power, trigger, echo)
- Frequency: 40Khz
- Model: HC-SR04



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Calculations

- Speed of sound in air = 343 m/s (approximately)
- 1m = 100 cm
- 1s = 1000 ms
- 1ms = 1000 micro s
- (343 meters/second) * (100 centimeters/meter) * (1 second/1000000 microseconds) = 0.0343 centimeters/microsecond
- The ultrasonic sensor measures the round-trip time of the sound wave. So divide by 2
- 0.0343 centimeters/microsecond / 2 = 0.01723 centimeters/microsecond

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Algorithm

- Set **Trigger Pin** as Output.
- **Clear Trigger:** digitalWrite(triggerPin, LOW): Sets the trigger pin to a low (0) state.
- **delayMicroseconds(2):** to ensure the trigger pin is stable before sending the pulse.
- **Send Trigger Pulse:** Sets the trigger pin to a high (1) state for 10 microseconds
- Set the **trigger pin** back to a **low** state.
- **Set Echo Pin as Input:** Configures the specified echo pin as an input pin.
- **Measure Echo Pulse Duration:** Uses the **pulseIn** function to measure the duration of the high pulse on the echo pin. The measured duration is **returned in microseconds.**
- **Show results in LCD.**

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pulseIn()

- **Purpose:** Measure duration between state change
- Syntax:

```
pulseIn(pin, value)

pulseIn(pin, value, timeout)
```

- Parameters:

  Pin: the number of the Arduino pin on which you want to read the pulse.

  Value: type of pulse to read: either HIGH or LOW

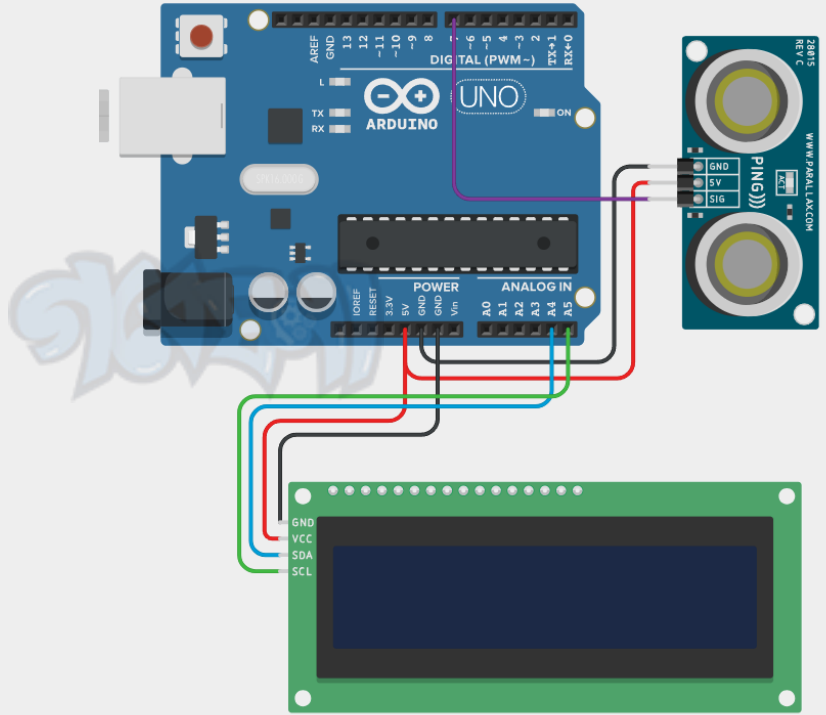
  Timeout (optional): the number of microseconds to wait for the pulse to start;

- **Returns:** The length of the pulse (in microseconds) or 0 if no pulse started before the timeout. Data type: unsigned long.

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Sample Circuit

- SDA: Analog 4 (A4)
- SCL Analog 5 (A5)
- Voltage level: 5V
- Signal pin: Digital 7



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Assignment

Write a code for measuring distance using sonar sensor.

Show results (in inch scale) on LCD display.

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7 Segment Display

- Usage: digit display
- Long lifetime
- Multi-segments

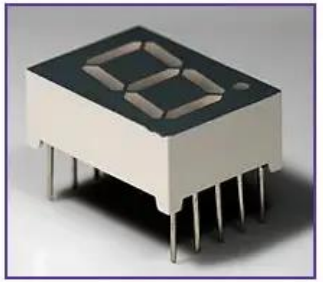


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7 Segment Display

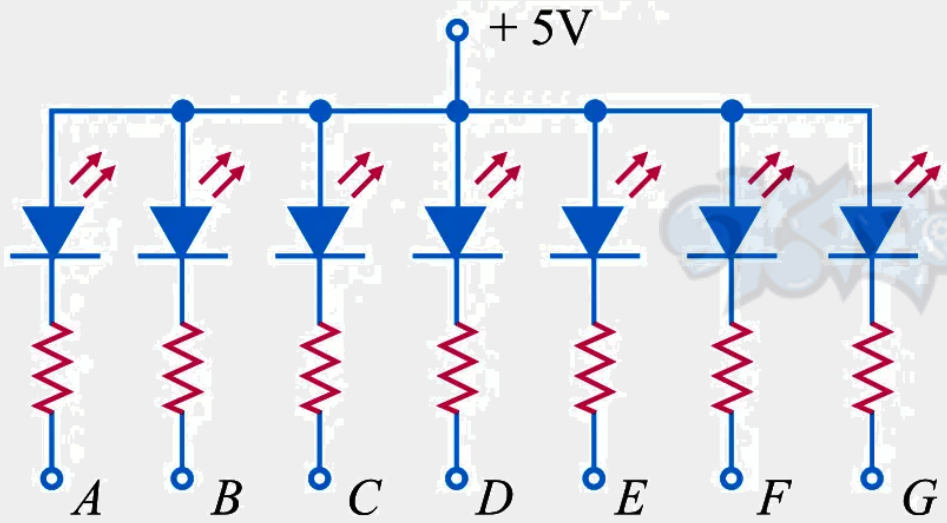
- 7 segments
- 2 types: Common anode, common cathode.
- Low current needed 30mA max
- Low Cost

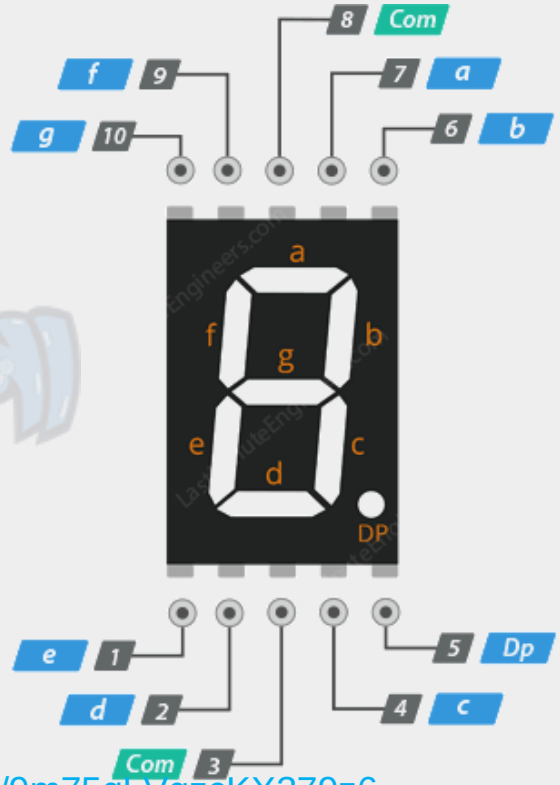




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Pinout

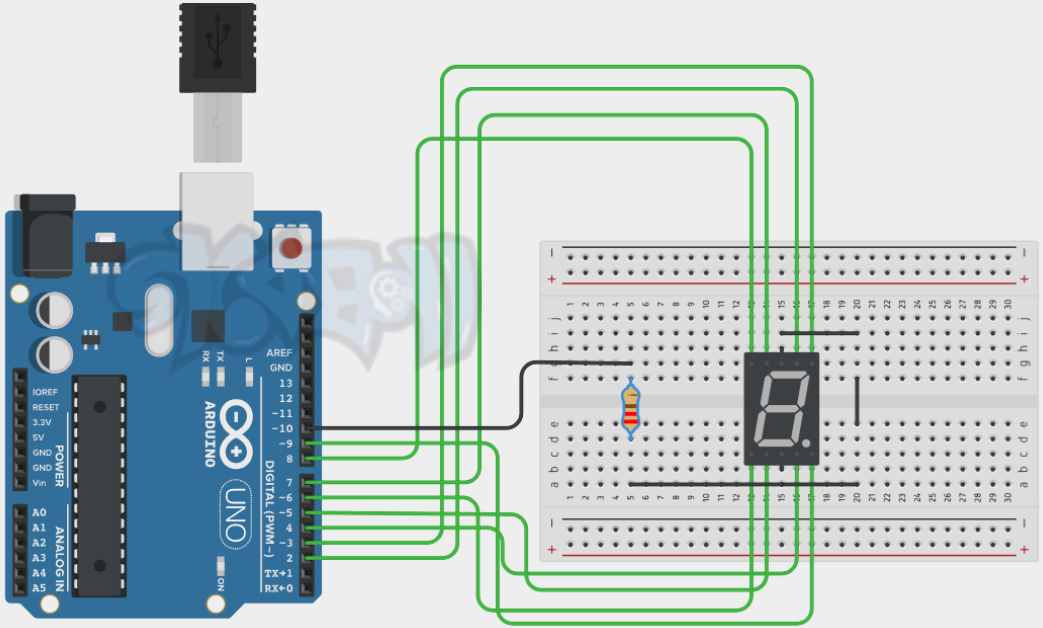




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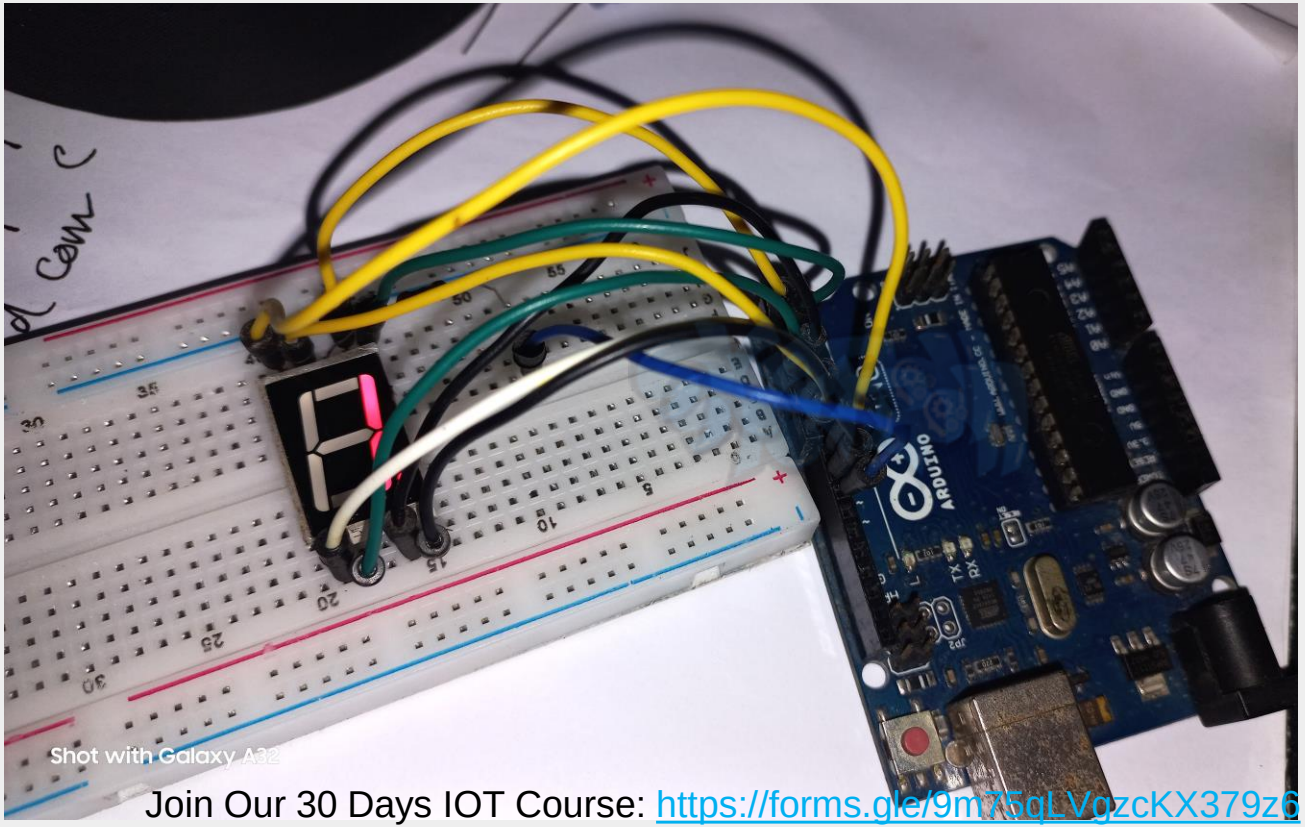
Sample Circuit

- A-G: 2-8
- DP: 9
- Common: 10



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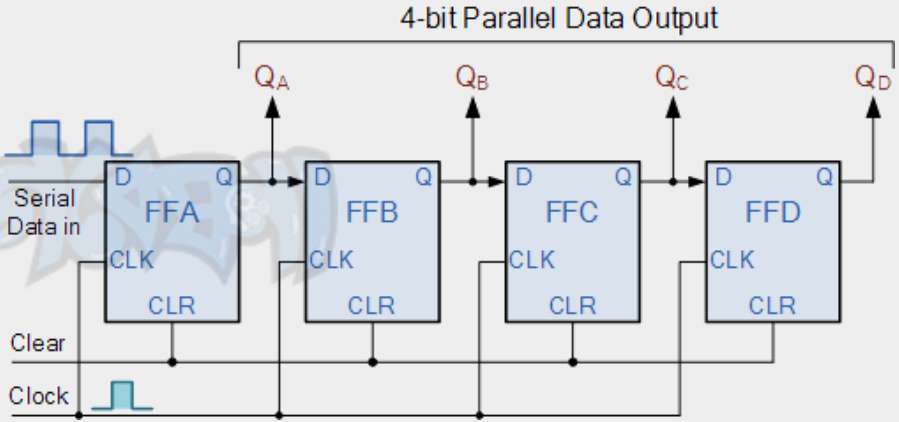
Sample Circuit



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Shift Register

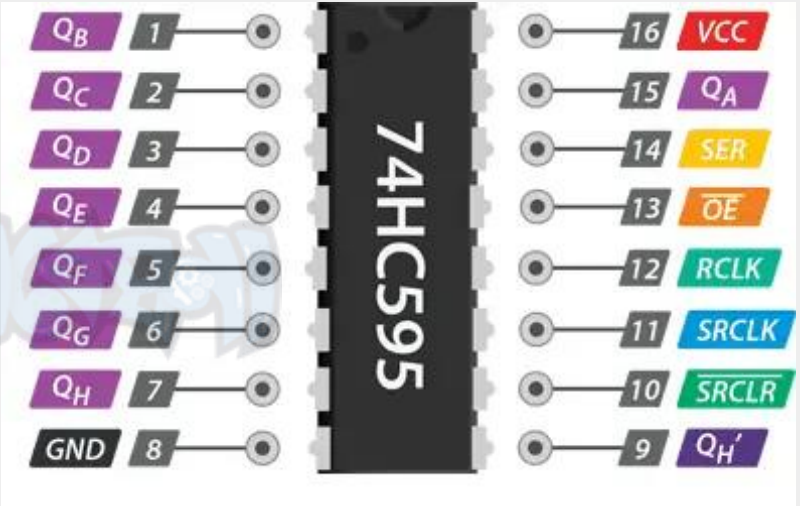
- Serial to Parallel converter
- 8 Output pins
- Only 3 pins needed to operate with Arduino



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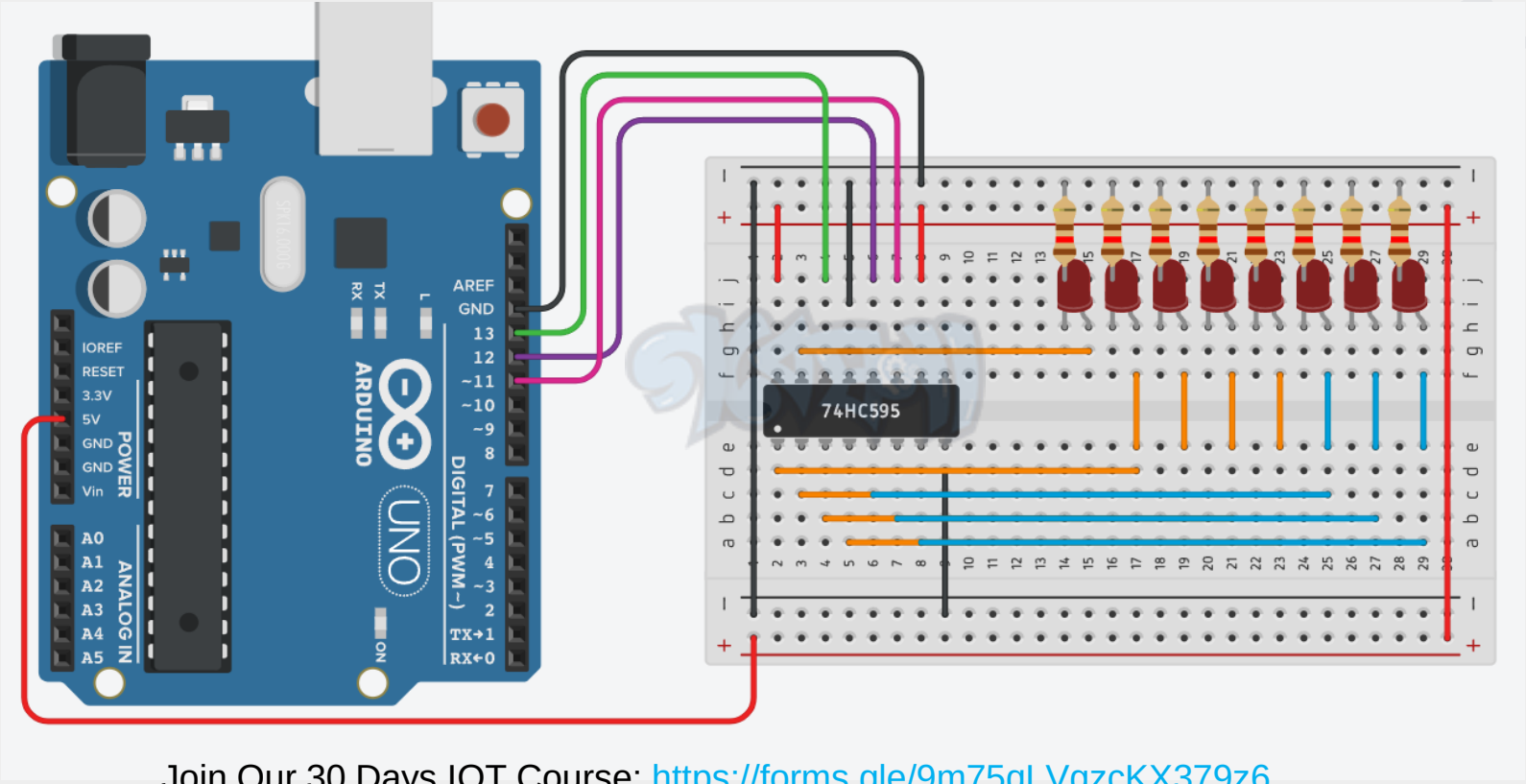
Pinout

- QA-QH: Parallel Out
- QH' : Serial Data out
- SRCLR' : Reset (5v)
- SRCLK: Clock
- RCLK: Latch
- OE' : Output Enable (GND)
- SER: Serial input



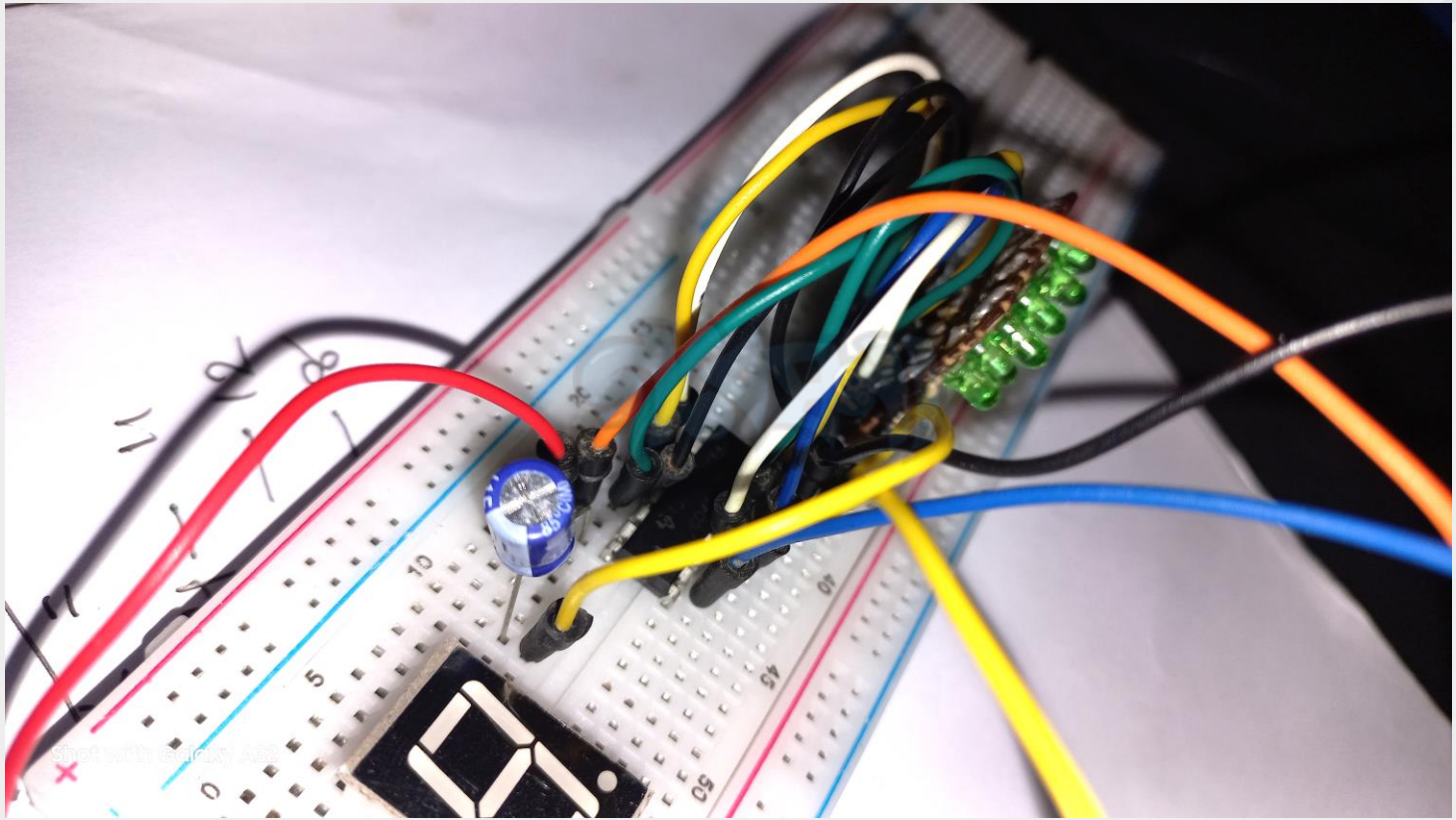
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Sample Circuit



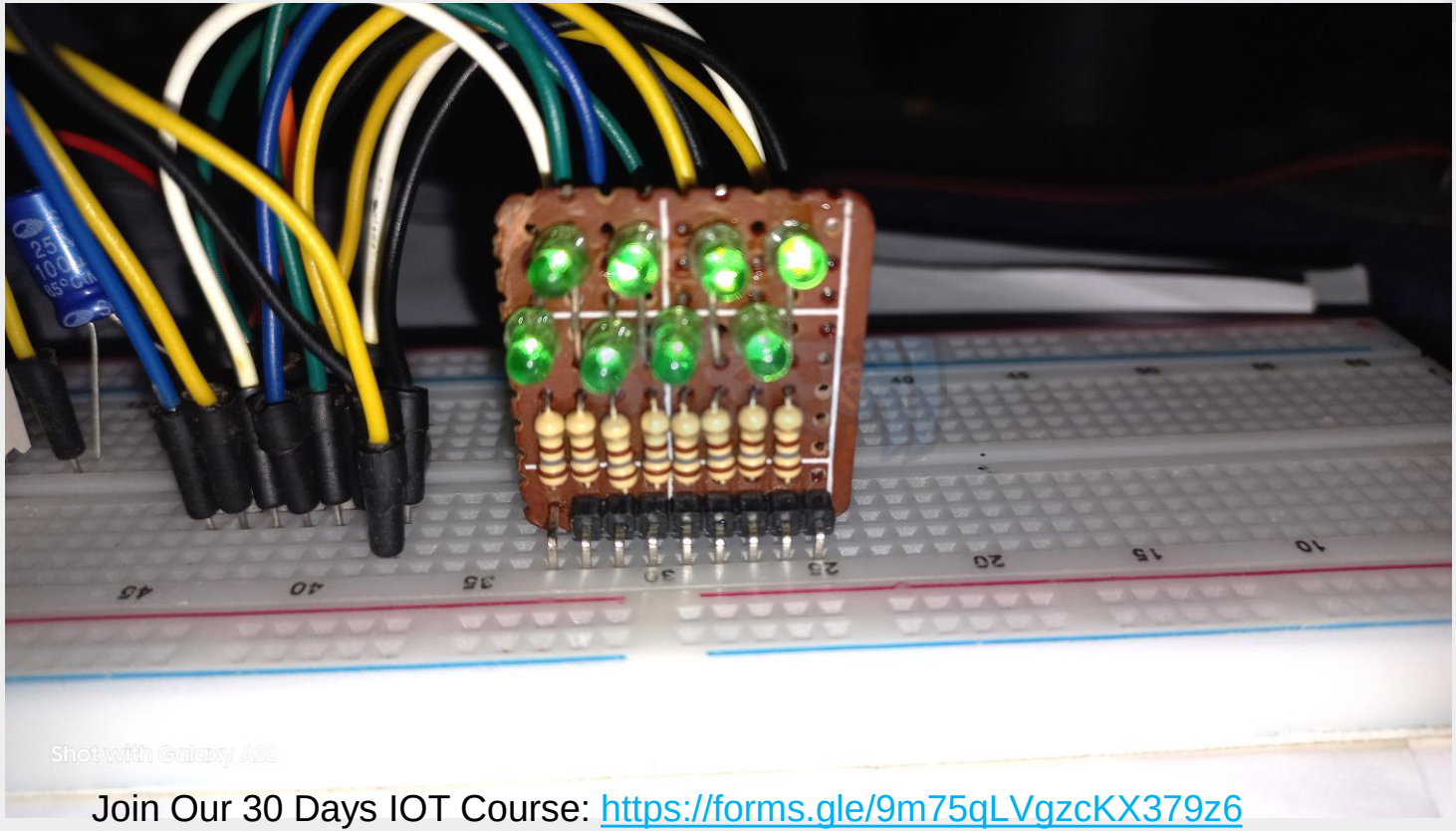
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Sample Circuit



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Sample Circuit



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Assignment

Write a code for 7 segment or 8 leds to show 0-9 or binary numbers

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Temperature Measurement

What is it?

- Temperature measures heat energy.
- Thermometers measure temperature.
- Common units are Celsius, Fahrenheit, and Kelvin.
- Analog and digital sensors measure temperature.
- Temperature affects many processes.

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* Image courtesy: Getty Images



DHT 11/12/22 Sensor

Features:

- Measure ambient temperature.
- Accuracy: High
- Fast speed to measure the value
- Also measures Humidity
- Digital 1 wire output

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* Image courtesy: Graylogix.in

Temperature Sense



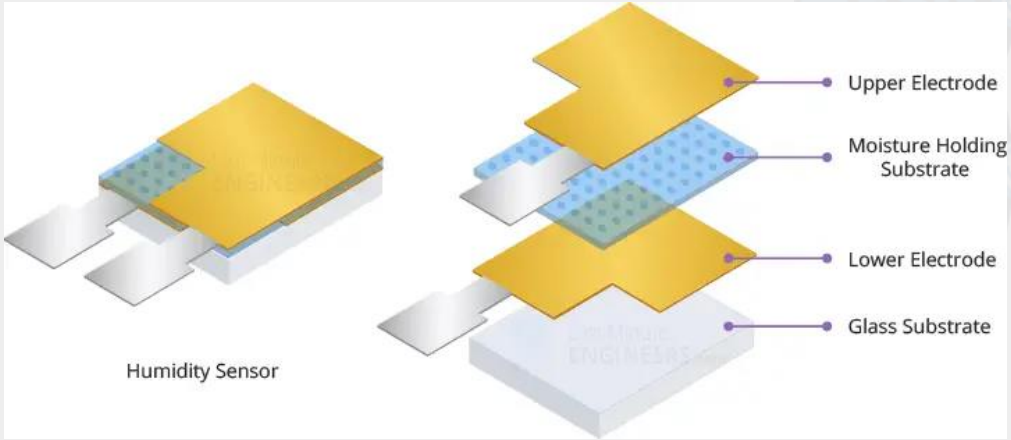
Features:

- Variable Resistor
- Resistance depends on temperature
- DHT sensor uses Thermistor (NTC)
- High accuracy
- Analog signal (output)

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* Image courtesy: jec1988.com

Humidity Sense



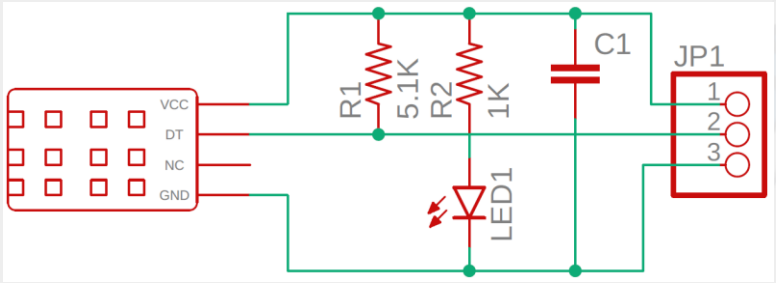
Features:

- DHT sensor uses Capacitive sensor
- Ceramic based capacitive particles
- High accuracy
- Analog signal

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* Image courtesy: viasala blog

Digital Conversion



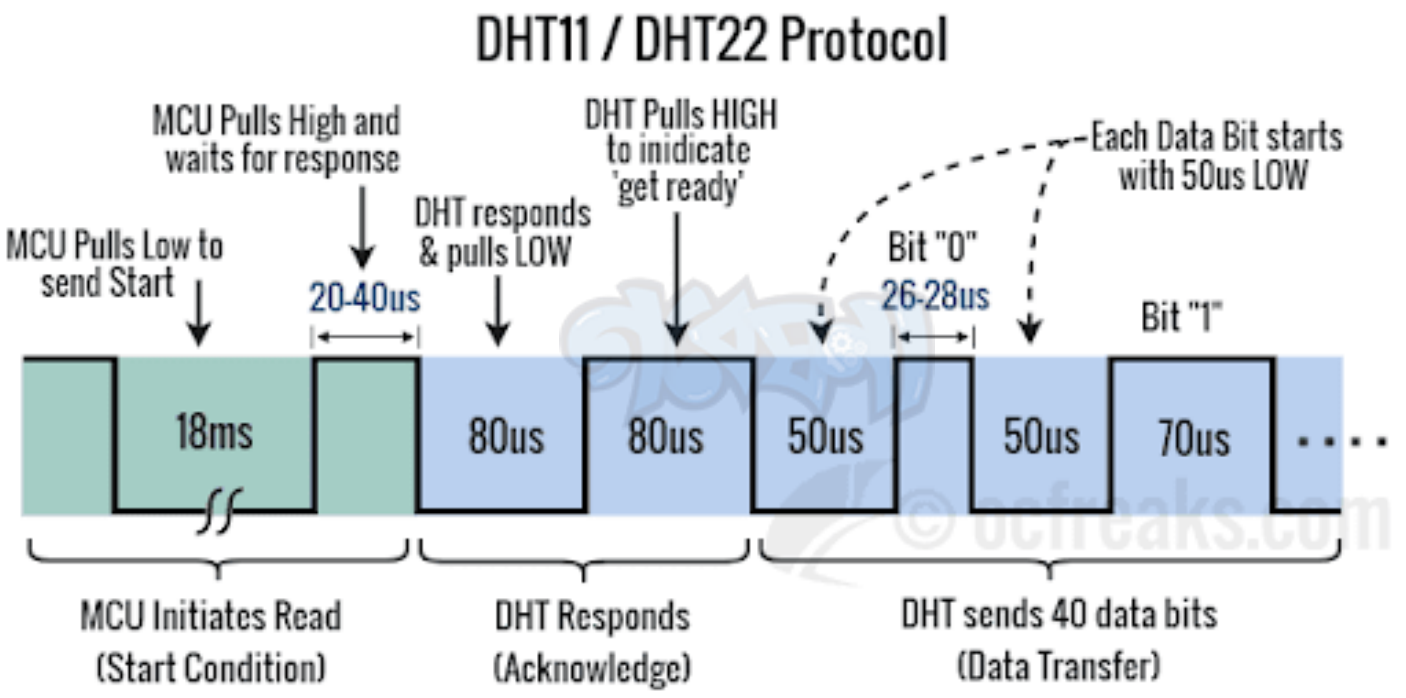
How?

- Uses an internal MCU to collect data
- Convert it to digital data
- Send data through 1 wire on request

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* Image courtesy: Easy EDA

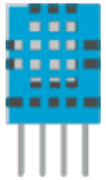
Timing Diagram

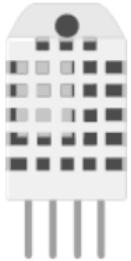


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* Image courtesy: ocfreaks.com

Different Versions



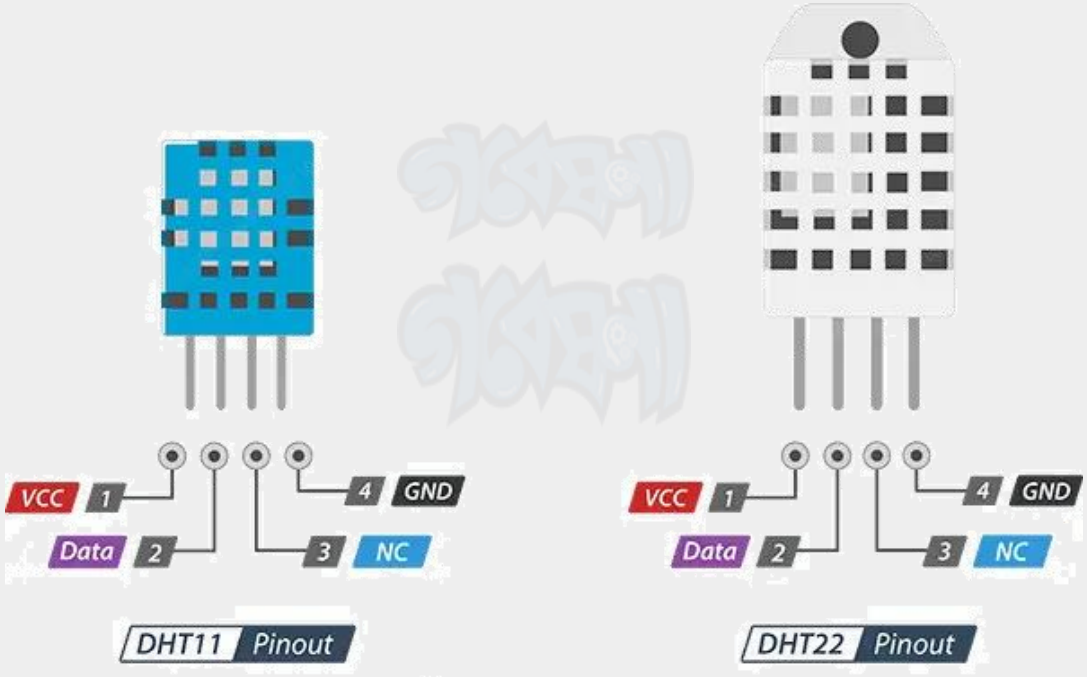


	DHT11	DHT22
Operating Voltage	3 to 5V	3 to 5V
Max Operating Current	2.5mA max	2.5mA max
Humidity Range	20-80% / 5%	0-100% / 2-5%
Temperature Range	0-50°C / ± 2°C	-40 to 80°C / ± 0.5°C
Sampling Rate	1 Hz (reading every second)	0.5 Hz (reading every 2 seconds)
Body size	15.5mm x 12mm x 5.5mm	15.1mm x 25mm x 7.7mm
Advantage	Ultra low cost	More Accurate

Image courtesy: Last minute Engineers

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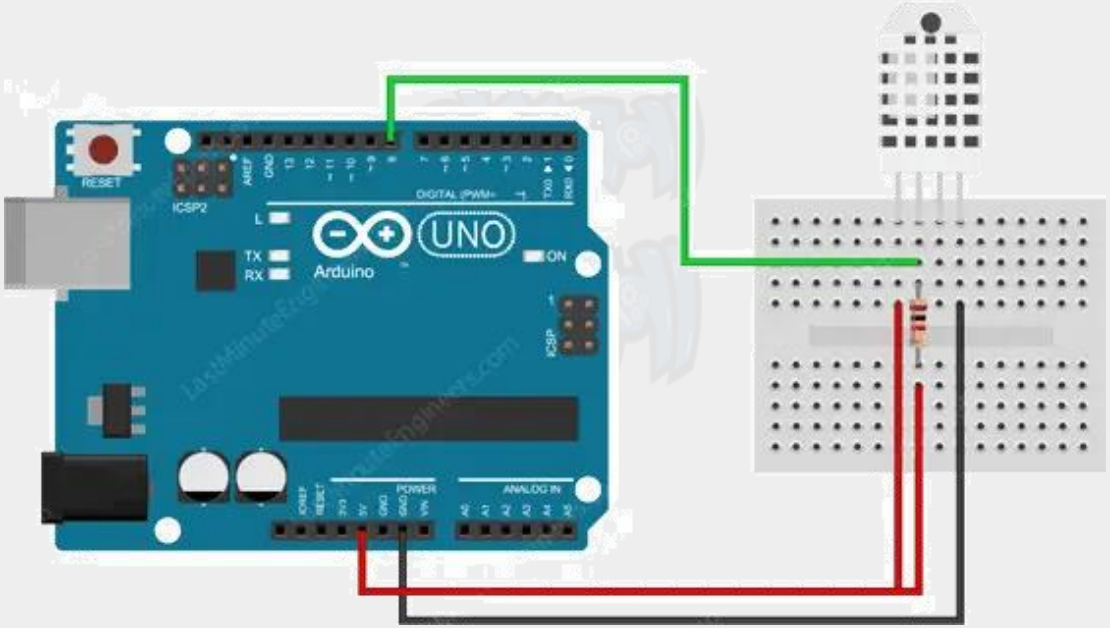
Pinouts



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* Image courtesy: Last minute Engineers

Sample Circuit



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* Image courtesy: Last minute Engineers



Assignment

Steps:

- Measure the temperature of the environment.
- If the temperature is more than 30, then turn on a LED.
- When the temperature is less than 20, then turn off LED.
- Submit video in Facebook group.

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* Image courtesy: iStock Photos

Any Questions?

Submit your assignment here:
https://www.facebook.com/groups/gobeshona

Thank you

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Keypad with Arduino

Required Features:

- Easy input from user
- No terminal required
- Fast response
- Low cost
- High endurance

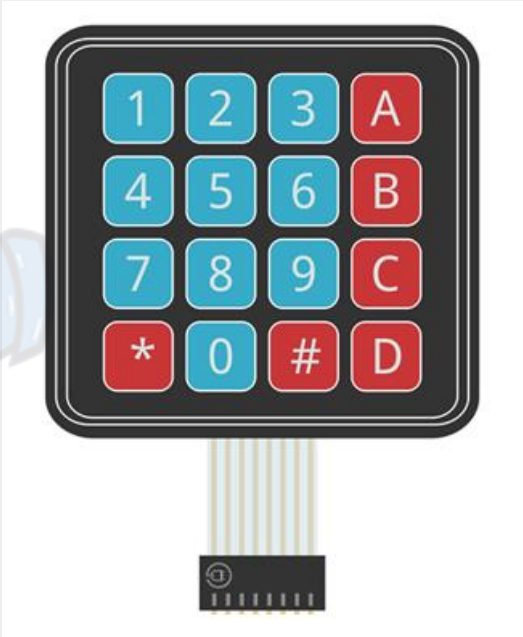


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4x4 Keypad

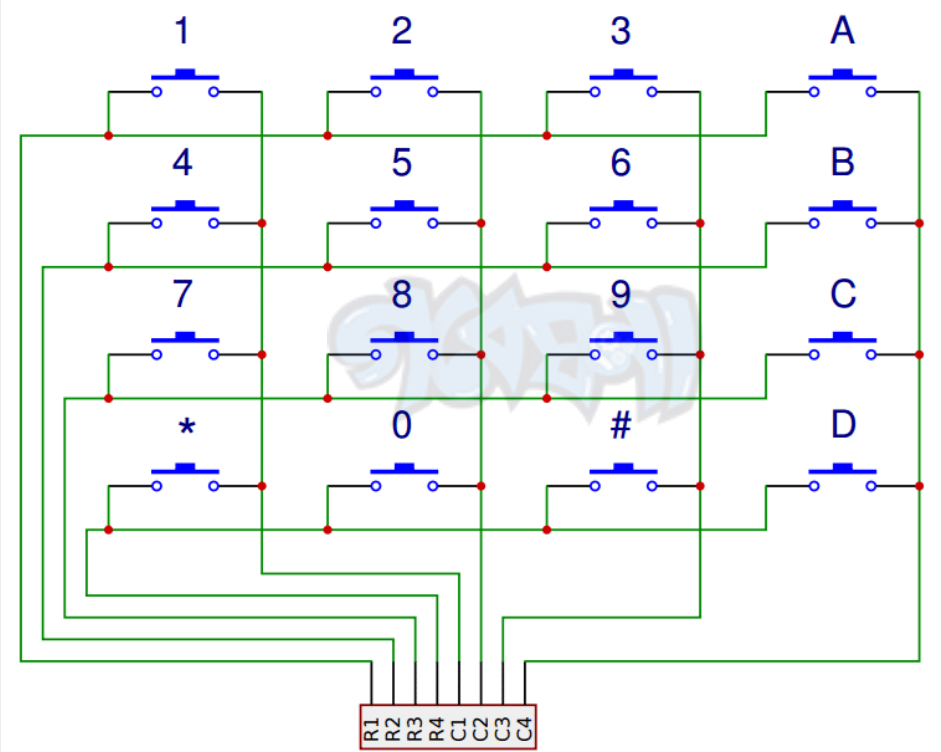
Features:

- Matrix/grid connection
- Promising performance
- 16 keys
- Low cost, durable
- Disadvantage: 8 digital pins required.



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Connection Diagram



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Rain/Moisture Sensor

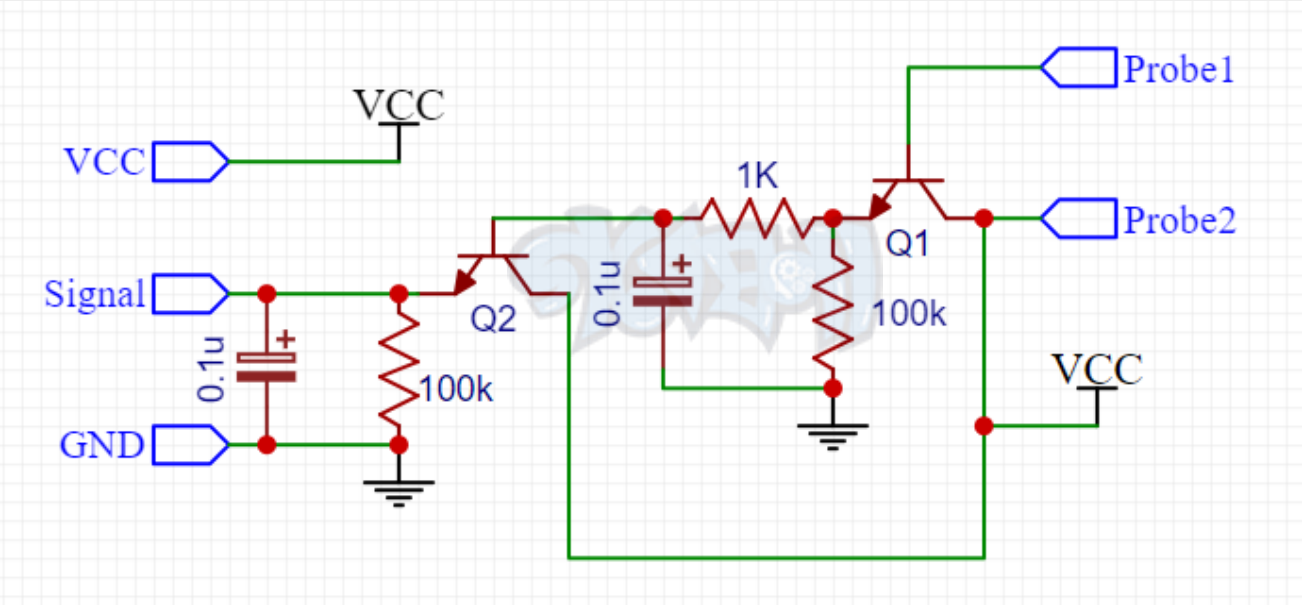
Features:

- Detects rain/water/moisture
- Low cost
- Easy to use



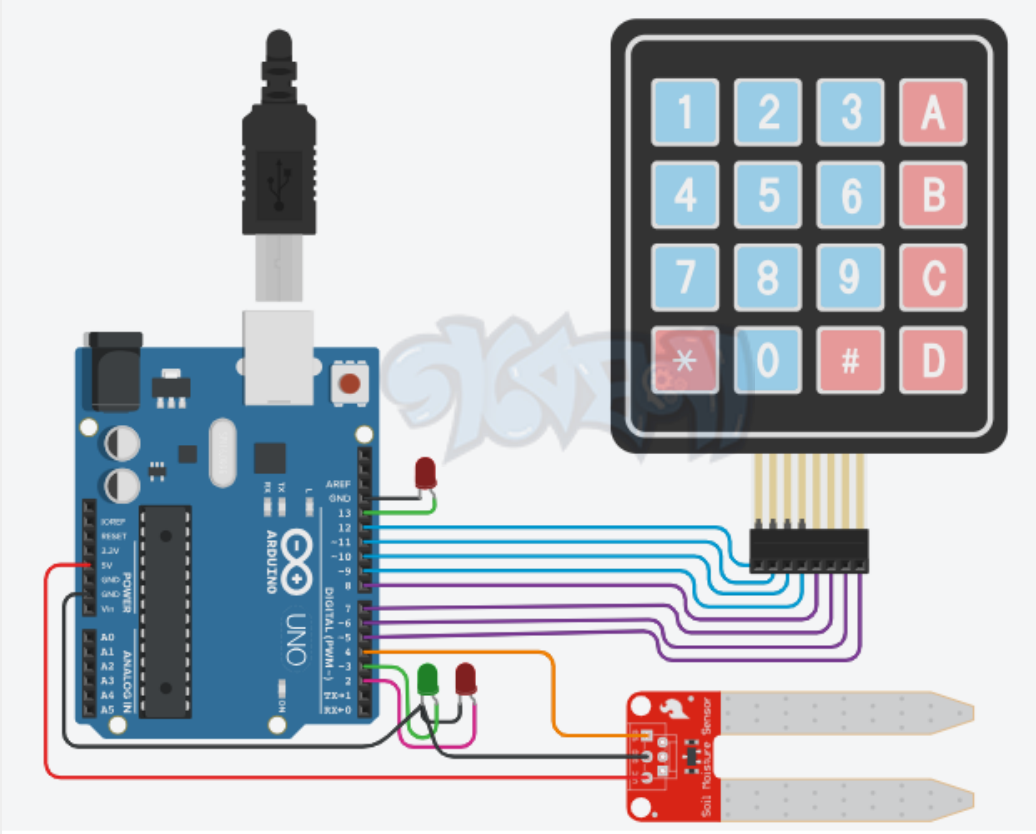
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Internal Diagram



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Sample Circuit



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Assignment

Measure the temperature of the environment. If the temperature is more than 30, then turn on a LED. When the temperature is less than 20, then turn off LED.

Use Keypad for manual override. Insert a passkey to turn on/off the system.

Passkey: 5678

Submit video in FACEBOOK group

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DC Motor

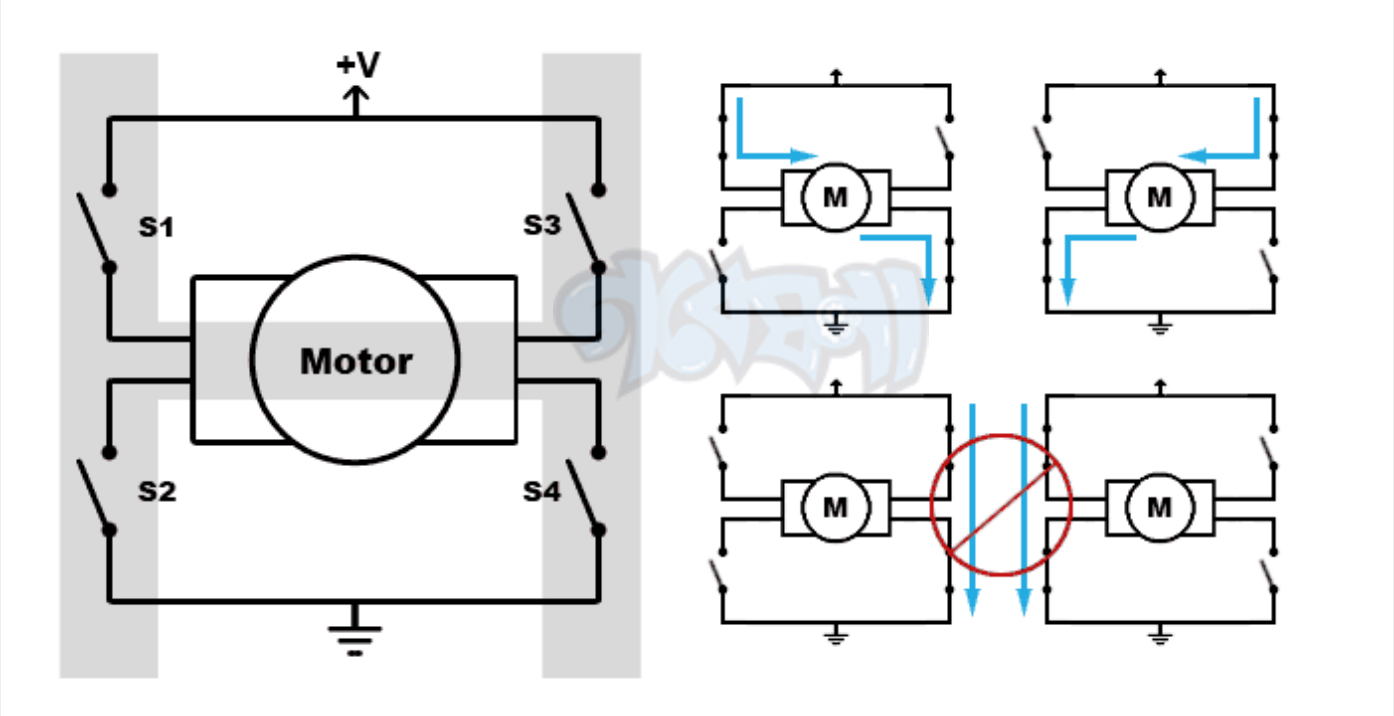
Required Features:

- Converts electrical energy to mechanical force.
- Reverse motion possible
- 2 wires
- High endurance
- Low cost



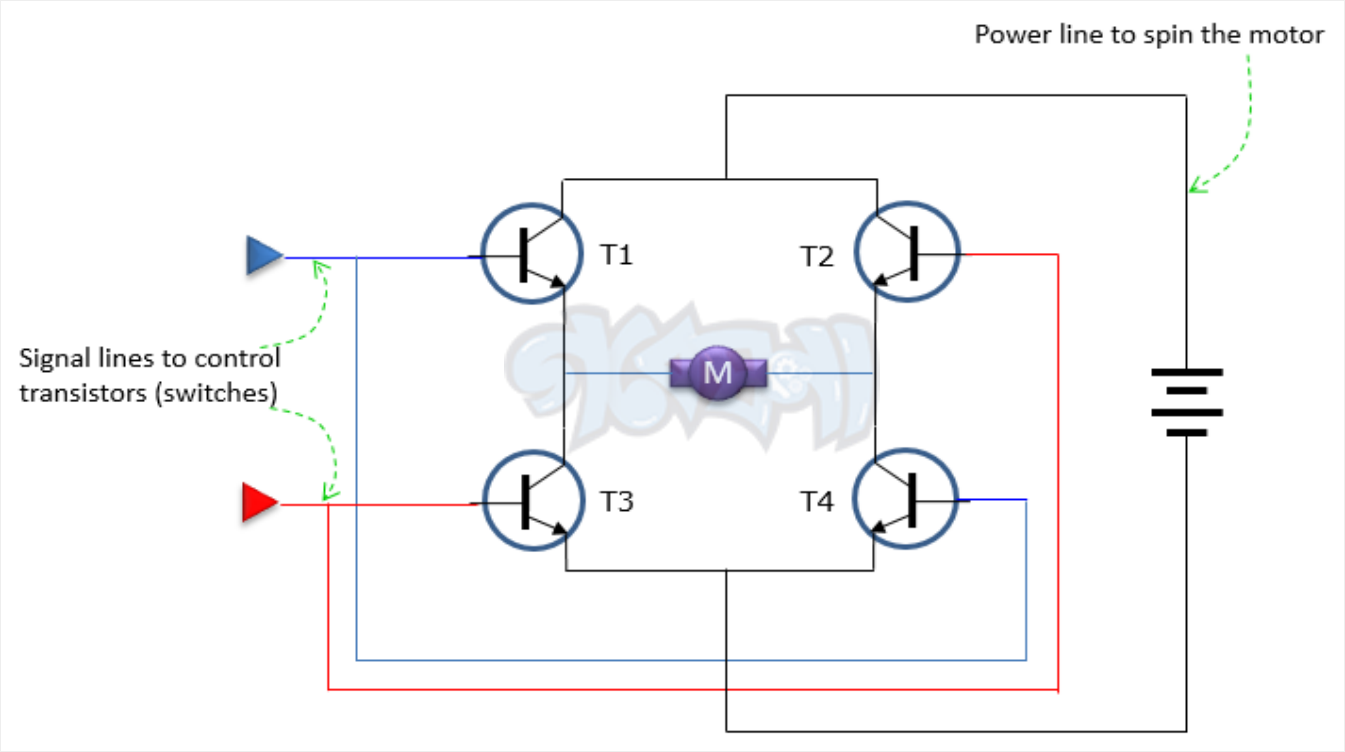
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Motor H-bridge



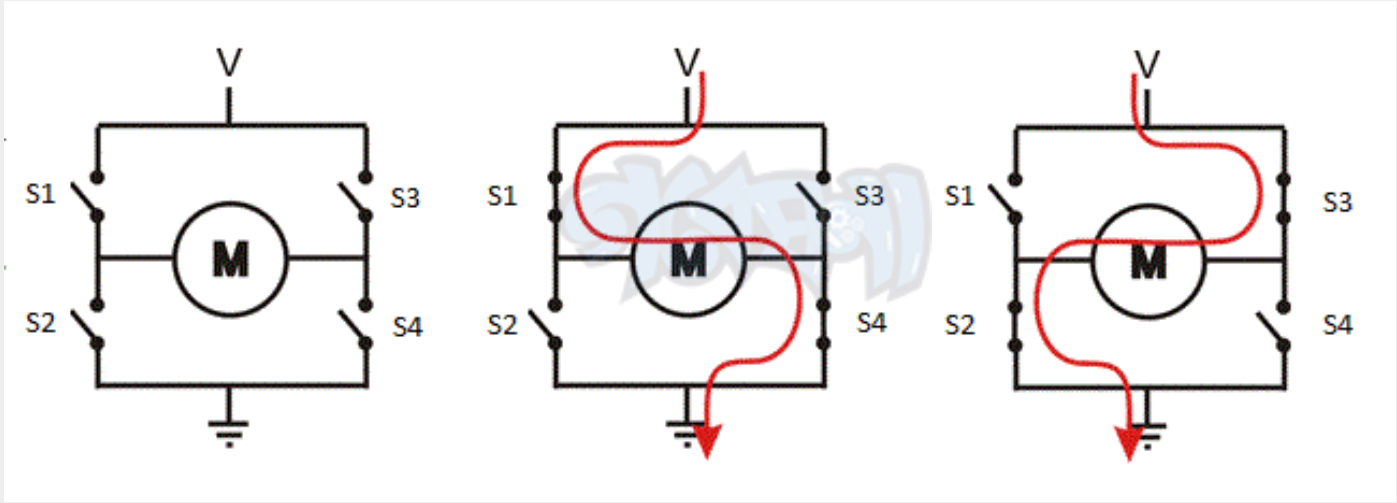
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Motor H-bridge



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Motor H-bridge

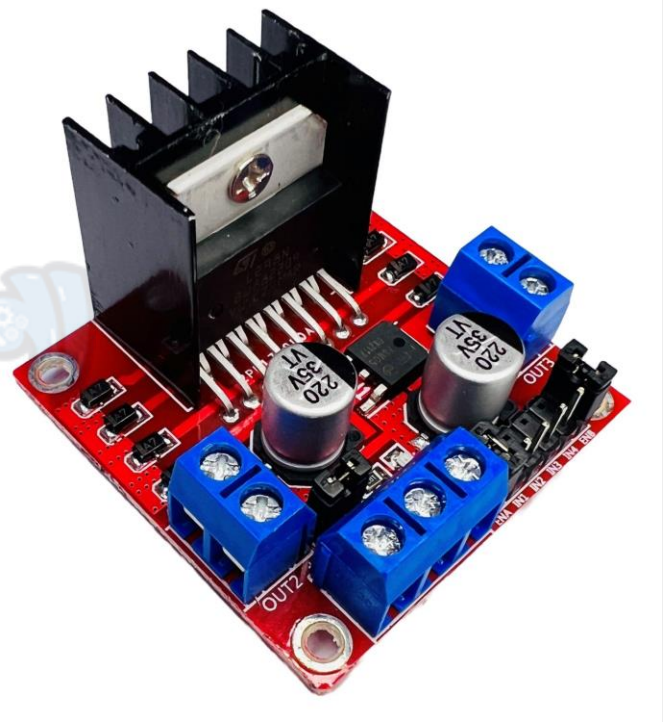


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DC Motor Driver

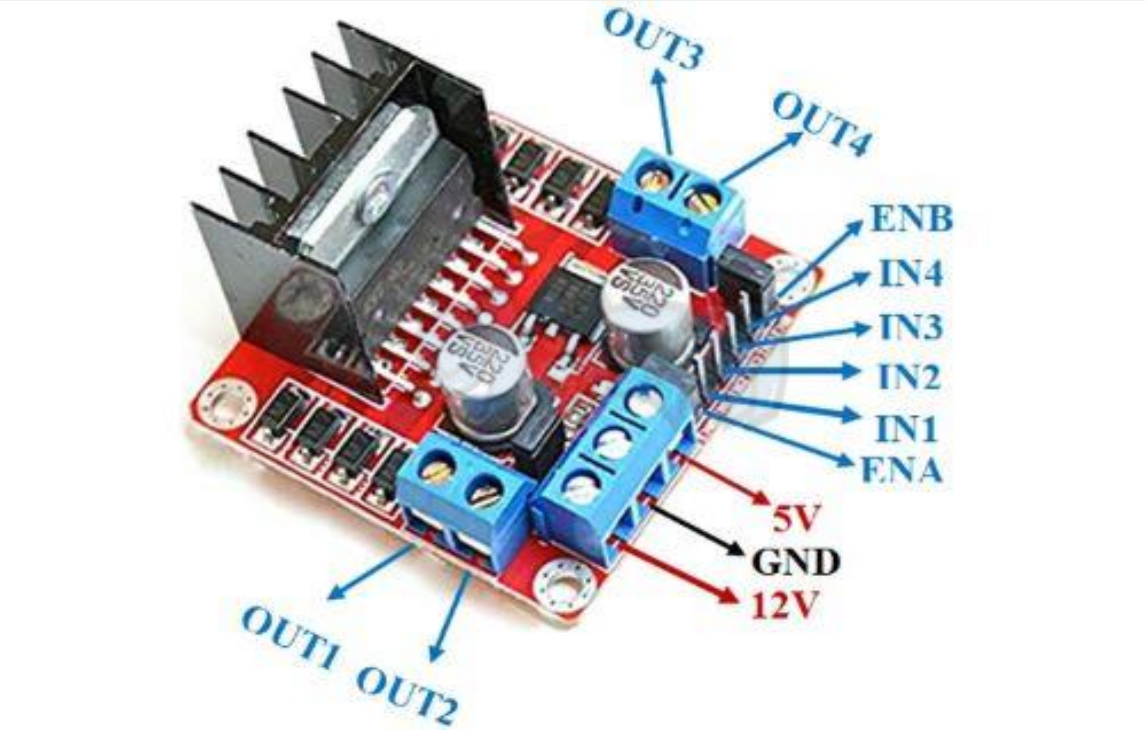
Features:

- Model: L298N
- Chip: Double H-bridge L298N
- Power: 25W
- Motor Supply Voltage: 46V
- Motor Supply Current: 2A
- Driver Voltage: 5-35V
- Driver Current: 2A



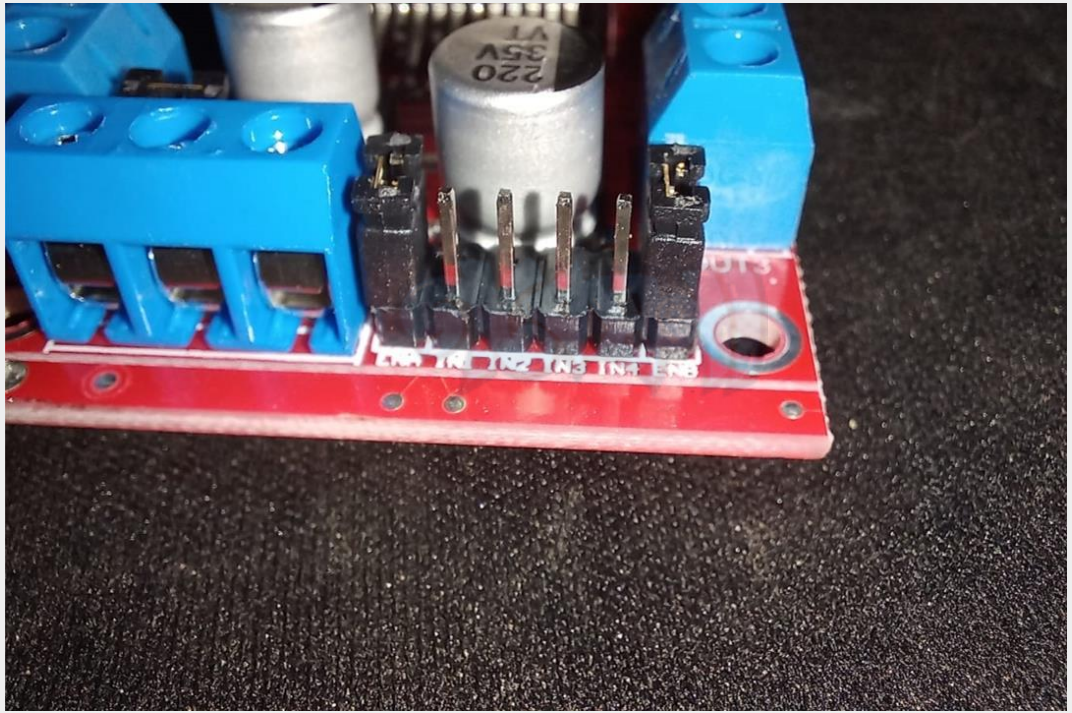
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Driver Pinout



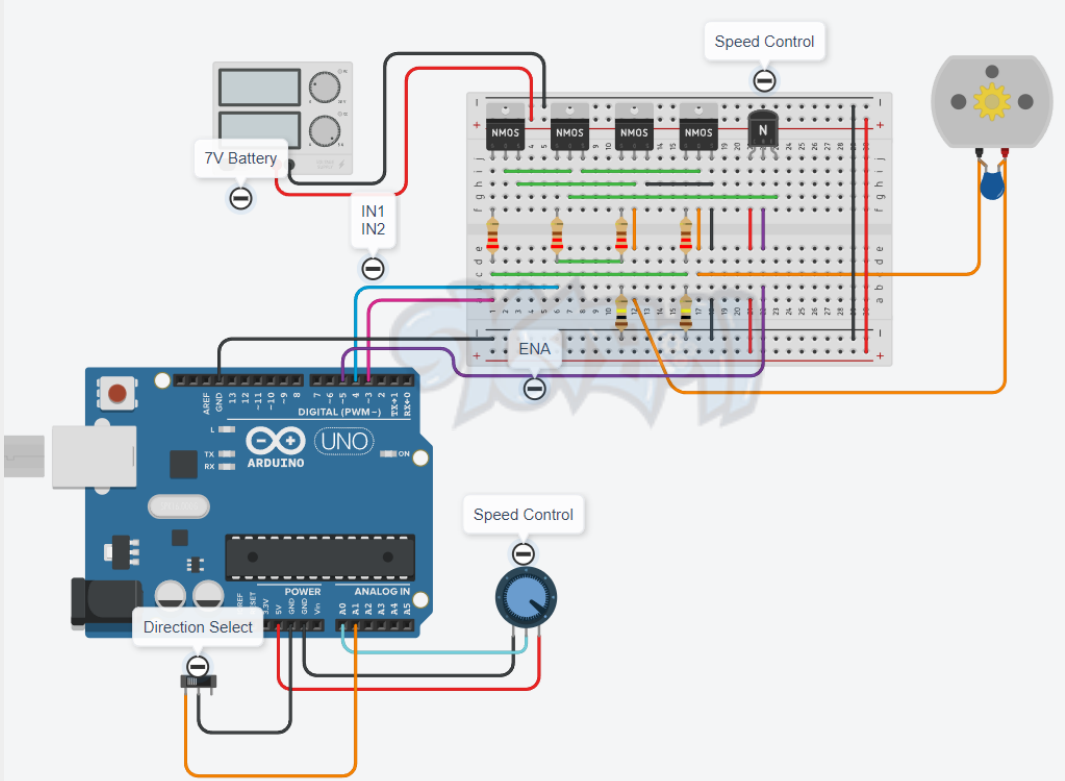
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Driver Pinout



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Sample Circuit



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Assignment

Control the speed of a motor using a motor driver. Add a switch to select the direction.

Also add a potentiometer to adjust the speed.

Submit video in FACEBOOK group

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